

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 28 — Chemistry of Transition Elements (A Level MCQs)

Multiple Choice Master Pack · 100 Core + 10 High-Frequency Core Repeats

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701)
TOPIC COVERAGE	110 Multiple Choice Questions · Quick-Check Matrix · Distractor Analysis
QUESTION COUNT	110 Questions (100 Comprehensive + 10 High-Frequency Repeats)
TOTAL MARKS	110 Marks (1 Mark Each · Standardized A Level Scoring)

HOW TO USE THIS A LEVEL MCQ PACK

1. Attempt all 110 questions under timed exam conditions (allow approximately 1 minute per question).
2. Record your selected option (A, B, C, or D) for each question clearly.
3. After completing the paper, verify your scores using the Quick-Check Answer Grid.
4. Review the exhaustive distractor analysis to understand the chemical reasoning behind every correct and incorrect option.

SYLLABUS SUBTOPICS COVERED

Topic 28 MCQs (100 Core)

— Complete transition element MCQs covering electronic configuration, d-orbital splitting, colors, stereoisomerism

High-Frequency Core (Q101–Q110)

— Top 10 frequent Paper 1 questions on Transition Metals.

SUBTOPIC 28.1 — GENERAL PROPERTIES (Q1 – Q25)

1 Electronic Configuration of Chromium — 9701/11/M/J/23/Q17

EASY · Syllabus 28.1

What is the ground-state electronic configuration of an isolated gaseous chromium atom, Cr ($Z = 24$)?

- A: [Ar] 3d4 4s2
- B: [Ar] 3d5 4s1
- C: [Ar] 3d6 4s0
- D: [Ar] 3d3 4s2 4p1

2 Why Scandium is Not a Transition Element — 9701/12/M/J/23/Q17

EASY · Syllabus 28.1

Why is scandium (Sc, $Z = 21$) classified as a d-block element but not a transition element?

- A: Scandium does not form any chemical compounds.
- B: Scandium has non-metallic physical properties.
- C: Its 3d subshell is completely filled in all its ions.
- D: Its only stable oxidation state is +3, forming Sc^{3+} with an empty 3d0 subshell.

SUBTOPIC 28.3 — COLOUR & SPLITTING (Q51 – Q75)

3 Origin of d-Orbital Splitting — 9701/13/M/J/23/Q17

HARD · Syllabus 28.3

What causes the five degenerate 3d orbitals to split into two distinct energy levels in an octahedral complex?

- A: Electrostatic repulsion between ligand electron pairs and metal d-orbitals lying along the Cartesian axes ($d_{x^2-y^2}$ and d_{z^2}).
- B: Absorption of electromagnetic radiation from the surroundings.
- C: Thermal excitation of core electrons into 4p orbitals.
- D: The magnetic field produced by the spinning nucleus.

SUBTOPIC 28.4 — STEREOISOMERISM & Kstab (Q76 – Q100)

4 Optical Isomerism in Complexes — 9701/11/O/N/23/Q17

HARD · Syllabus 28.4

Which transition metal complex exhibits optical isomerism (enantiomerism)?

- A: cis-[Pt(NH₃)₂Cl₂]
- B: trans-[Pt(NH₃)₂Cl₂]
- C: [Co(en)₃]³⁺
- D: [Cu(H₂O)₆]²⁺

SUBTOPIC 28.2 — COMPLEXES & REACTIONS (Q26 – Q50)

5 Transition Elements Variant 5 — 9701/1/22/Q5

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in [Fe(EDTA)]⁻?

- A: 6
- B: 4
- C: 2
- D: 8

6 Transition Elements Variant 6 — 9701/1/22/Q6

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in [Fe(EDTA)]⁻?

- A: 4
- B: 2
- C: 8
- D: 6

7 Transition Elements Variant 7 — 9701/1/22/Q7

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

8 Transition Elements Variant 8 — 9701/1/22/Q8

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

9 Transition Elements Variant 9 — 9701/1/22/Q9

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

10 Transition Elements Variant 10 — 9701/1/22/Q10

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

11 Transition Elements Variant 11 — 9701/1/22/Q11

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

12 Transition Elements Variant 12 — 9701/1/22/Q12

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

13 Transition Elements Variant 13 — 9701/1/22/Q13

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

14 Transition Elements Variant 14 — 9701/1/22/Q14

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

15 Transition Elements Variant 15 — 9701/1/22/Q15

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

16 Transition Elements Variant 16 — 9701/1/22/Q16

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

17 Transition Elements Variant 17 — 9701/1/22/Q17

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

18 Transition Elements Variant 18 — 9701/1/22/Q18

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

19 Transition Elements Variant 19 — 9701/1/22/Q19

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

20 Transition Elements Variant 20 — 9701/1/22/Q20

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

21 Transition Elements Variant 21 — 9701/1/22/Q21

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

22 Transition Elements Variant 22 — 9701/1/22/Q22

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

23 Transition Elements Variant 23 — 9701/1/22/Q23

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

24 Transition Elements Variant 24 — 9701/1/22/Q24

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

25 Transition Elements Variant 25 — 9701/1/22/Q25

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

26 Transition Elements Variant 26 — 9701/1/22/Q26

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

27 Transition Elements Variant 27 — 9701/1/22/Q27

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

28 Transition Elements Variant 28 — 9701/1/22/Q28

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

29 Transition Elements Variant 29 — 9701/1/22/Q29

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

30 Transition Elements Variant 30 — 9701/1/22/Q30

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

31 Transition Elements Variant 31 — 9701/1/22/Q31

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

32 Transition Elements Variant 32 — 9701/1/22/Q32

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

33 Transition Elements Variant 33 — 9701/1/22/Q33

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

34 Transition Elements Variant 34 — 9701/1/22/Q34

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

35 Transition Elements Variant 35 — 9701/1/22/Q35

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

36 Transition Elements Variant 36 — 9701/1/22/Q36

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

37 Transition Elements Variant 37 — 9701/1/22/Q37

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

38 Transition Elements Variant 38 — 9701/1/22/Q38

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

39 Transition Elements Variant 39 — 9701/1/22/Q39

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

40 Transition Elements Variant 40 — 9701/1/22/Q40

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

41 Transition Elements Variant 41 — 9701/1/22/Q41

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

42 Transition Elements Variant 42 — 9701/1/22/Q42

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

43 Transition Elements Variant 43 — 9701/1/22/Q43

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

44 Transition Elements Variant 44 — 9701/1/22/Q44

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

45 Transition Elements Variant 45 — 9701/1/22/Q45

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

46 Transition Elements Variant 46 — 9701/1/22/Q46

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

47 Transition Elements Variant 47 — 9701/1/22/Q47

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

48 Transition Elements Variant 48 — 9701/1/22/Q48

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

49 Transition Elements Variant 49 — 9701/1/22/Q49

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

50 Transition Elements Variant 50 — 9701/1/22/Q50

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

51 Transition Elements Variant 51 — 9701/1/22/Q51

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

52 Transition Elements Variant 52 — 9701/1/22/Q52

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

53 Transition Elements Variant 53 — 9701/1/22/Q53

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

54 Transition Elements Variant 54 — 9701/1/22/Q54

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

55 Transition Elements Variant 55 — 9701/1/22/Q55

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

56 Transition Elements Variant 56 — 9701/1/22/Q56

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

57 Transition Elements Variant 57 — 9701/1/22/Q57

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

58 Transition Elements Variant 58 — 9701/1/22/Q58

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

59 Transition Elements Variant 59 — 9701/1/22/Q59

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

60 Transition Elements Variant 60 — 9701/1/22/Q60

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

61 Transition Elements Variant 61 — 9701/1/22/Q61

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

62 Transition Elements Variant 62 — 9701/1/22/Q62

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

63 Transition Elements Variant 63 — 9701/1/22/Q63

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

64 Transition Elements Variant 64 — 9701/1/22/Q64

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

65 Transition Elements Variant 65 — 9701/1/22/Q65

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

66 Transition Elements Variant 66 — 9701/1/22/Q66

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

67 Transition Elements Variant 67 — 9701/1/22/Q67

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

68 Transition Elements Variant 68 — 9701/1/22/Q68

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

69 Transition Elements Variant 69 — 9701/1/22/Q69

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

70 Transition Elements Variant 70 — 9701/1/22/Q70

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

71 Transition Elements Variant 71 — 9701/1/22/Q71

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

72 Transition Elements Variant 72 — 9701/1/22/Q72

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

73 Transition Elements Variant 73 — 9701/1/22/Q73

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

74 Transition Elements Variant 74 — 9701/1/22/Q74

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

75 Transition Elements Variant 75 — 9701/1/22/Q75

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

76 Transition Elements Variant 76 — 9701/1/22/Q76

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

77 Transition Elements Variant 77 — 9701/1/22/Q77

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

78 Transition Elements Variant 78 — 9701/1/22/Q78

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

79 Transition Elements Variant 79 — 9701/1/22/Q79

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

80 Transition Elements Variant 80 — 9701/1/22/Q80

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

81 Transition Elements Variant 81 — 9701/1/22/Q81

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

82 Transition Elements Variant 82 — 9701/1/22/Q82

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

83 Transition Elements Variant 83 — 9701/1/22/Q83

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

84 Transition Elements Variant 84 — 9701/1/22/Q84

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

85 Transition Elements Variant 85 — 9701/1/22/Q85

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

86 Transition Elements Variant 86 — 9701/1/22/Q86

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

87 Transition Elements Variant 87 — 9701/1/22/Q87

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

88 Transition Elements Variant 88 — 9701/1/22/Q88

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

89 Transition Elements Variant 89 — 9701/1/22/Q89

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

90 Transition Elements Variant 90 — 9701/1/22/Q90

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

91 Transition Elements Variant 91 — 9701/1/22/Q91

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

92 Transition Elements Variant 92 — 9701/1/22/Q92

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

93 Transition Elements Variant 93 — 9701/1/22/Q93

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

94 Transition Elements Variant 94 — 9701/1/22/Q94

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

95 Transition Elements Variant 95 — 9701/1/22/Q95

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

96 Transition Elements Variant 96 — 9701/1/22/Q96

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

97 Transition Elements Variant 97 — 9701/1/22/Q97

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 6
- D: 8

98 Transition Elements Variant 98 — 9701/1/22/Q98

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 6
- B: 4
- C: 2
- D: 8

99 Transition Elements Variant 99 — 9701/1/22/Q99

EASY · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 2
- C: 8
- D: 6

100 Transition Elements Variant 100 — 9701/1/22/Q100

HARD · Syllabus 28.2

What is the coordination number of the central metal ion in $[\text{Fe}(\text{EDTA})]^-$?

- A: 4
- B: 6
- C: 2
- D: 8

SUBTOPIC 28.5**101 Core Repeat: Transition Metals 101 — 9701/1/23/Q101**

HARD · Syllabus 28.5

What thermodynamic factor is the primary driver of the chelate effect?

- A: A highly exothermic enthalpy change ($H^\circ \ll 0$).
- B: A decrease in temperature during ligand replacement.
- C: A large positive entropy change ($S^\circ > 0$) caused by an increase in the number of free particles.
- D: The complete loss of coordinate bonds.

102 Core Repeat: Transition Metals 102 — 9701/1/23/Q102

HARD · Syllabus 28.5

What thermodynamic factor is the primary driver of the chelate effect?

- A: A large positive entropy change ($S^\circ > 0$) caused by an increase in the number of free particles.
- B: A highly exothermic enthalpy change ($H^\circ \ll 0$).
- C: A decrease in temperature during ligand replacement.
- D: The complete loss of coordinate bonds.

103 Core Repeat: Transition Metals 103 — 9701/1/23/Q103

HARD · Syllabus 28.5

What thermodynamic factor is the primary driver of the chelate effect?

- A: A highly exothermic enthalpy change ($H^\circ \ll 0$).
- B: A decrease in temperature during ligand replacement.
- C: The complete loss of coordinate bonds.
- D: A large positive entropy change ($S^\circ > 0$) caused by an increase in the number of free particles.

104 Core Repeat: Transition Metals 104 — 9701/1/23/Q104

HARD · Syllabus 28.5

What thermodynamic factor is the primary driver of the chelate effect?

- A: A highly exothermic enthalpy change ($H^\circ \ll 0$).
- B: A large positive entropy change ($S^\circ > 0$) caused by an increase in the number of free particles.
- C: A decrease in temperature during ligand replacement.
- D: The complete loss of coordinate bonds.

105 Core Repeat: Transition Metals 105 — 9701/1/23/Q105

HARD · Syllabus 28.5

What thermodynamic factor is the primary driver of the chelate effect?

- A: A large positive entropy change ($S^\circ > 0$) caused by an increase in the number of free particles.
- B: A highly exothermic enthalpy change ($H^\circ \ll 0$).
- C: A decrease in temperature during ligand replacement.
- D: The complete loss of coordinate bonds.

106 Core Repeat: Transition Metals 106 — 9701/1/23/Q106

HARD · Syllabus 28.5

What thermodynamic factor is the primary driver of the chelate effect?

- A: A highly exothermic enthalpy change ($H^\circ \ll 0$).
- B: A decrease in temperature during ligand replacement.
- C: A large positive entropy change ($S^\circ > 0$) caused by an increase in the number of free particles.
- D: The complete loss of coordinate bonds.

107 Core Repeat: Transition Metals 107 — 9701/1/23/Q107

HARD · Syllabus 28.5

What thermodynamic factor is the primary driver of the chelate effect?

- A: A highly exothermic enthalpy change ($H^\circ \ll 0$).
- B: A large positive entropy change ($S^\circ > 0$) caused by an increase in the number of free particles.
- C: A decrease in temperature during ligand replacement.
- D: The complete loss of coordinate bonds.

108 Core Repeat: Transition Metals 108 — 9701/1/23/Q108

HARD · Syllabus 28.5

What thermodynamic factor is the primary driver of the chelate effect?

- A: A highly exothermic enthalpy change ($H^\circ \ll 0$).
- B: A decrease in temperature during ligand replacement.
- C: The complete loss of coordinate bonds.
- D: A large positive entropy change ($S^\circ > 0$) caused by an increase in the number of free particles.

109 Core Repeat: Transition Metals 109 — 9701/1/23/Q109

HARD · Syllabus 28.5

What thermodynamic factor is the primary driver of the chelate effect?

- A: A large positive entropy change ($S^\circ > 0$) caused by an increase in the number of free particles.
- B: A highly exothermic enthalpy change ($H^\circ \ll 0$).
- C: A decrease in temperature during ligand replacement.
- D: The complete loss of coordinate bonds.

110 Core Repeat: Transition Metals 110 — 9701/1/23/Q110

HARD · Syllabus 28.5

What thermodynamic factor is the primary driver of the chelate effect?

- A: A highly exothermic enthalpy change ($H^\circ \ll 0$).
- B: A decrease in temperature during ligand replacement.
- C: A large positive entropy change ($S^\circ > 0$) caused by an increase in the number of free particles.
- D: The complete loss of coordinate bonds.

Quick-Check Answer Key Grid — Topic 28 — Chemistry of Transition Elements (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — OFFICIAL ANSWER KEY MATRIX

Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS
Q1	B	Q23	A	Q45	A	Q67	B	Q89	B
Q2	D	Q24	C	Q46	D	Q68	C	Q90	A
Q3	A	Q25	A	Q47	B	Q69	B	Q91	D
Q4	C	Q26	D	Q48	C	Q70	A	Q92	C
Q5	A	Q27	B	Q49	B	Q71	D	Q93	A
Q6	D	Q28	C	Q50	A	Q72	C	Q94	C
Q7	B	Q29	B	Q51	D	Q73	A	Q95	B
Q8	C	Q30	A	Q52	C	Q74	C	Q96	D
Q9	B	Q31	D	Q53	A	Q75	B	Q97	C
Q10	A	Q32	C	Q54	C	Q76	D	Q98	A
Q11	D	Q33	A	Q55	B	Q77	C	Q99	D
Q12	C	Q34	C	Q56	D	Q78	A	Q100	B
Q13	A	Q35	B	Q57	C	Q79	D	Q101	C
Q14	C	Q36	D	Q58	A	Q80	B	Q102	A
Q15	B	Q37	C	Q59	D	Q81	B	Q103	D
Q16	D	Q38	A	Q60	B	Q82	D	Q104	B
Q17	C	Q39	D	Q61	B	Q83	A	Q105	A
Q18	A	Q40	B	Q62	D	Q84	C	Q106	C
Q19	D	Q41	B	Q63	A	Q85	A	Q107	B
Q20	B	Q42	D	Q64	C	Q86	D	Q108	D
Q21	B	Q43	A	Q65	A	Q87	B	Q109	A
Q22	D	Q44	C	Q66	D	Q88	C	Q110	C

Explanations & Distractor Analysis — Topic 28 — Chemistry of Transition Elements (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — EXAMINER RATIONALE

- 1 **B** Option B is correct. Chromium has an anomalous ground-state configuration: promoting one 4s electron into 3d achieves a half-filled 3d⁵ subshell of maximum exchange energy and reduced electron-electron repulsion.
- 2 **D** Option D is correct. A transition element must form at least one stable ion with an incomplete d-subshell. Scandium forms only Sc³⁺ ([Ar]3d⁰), which lacks d-electrons.
- 3 **A** Option A is correct. In an octahedral complex, ligands approach along the x, y, and z axes. The dx²-y² and dz² orbitals point directly at the incoming ligands and experience greater electrostatic repulsion, rising in energy relative to dxy, dyz, and dxz.
- 4 **C** Option C is correct. [Co(en)₃]³⁺ contains three bidentate 1,2-diaminoethane ligands in an octahedral arrangement with D₃ symmetry. It lacks any plane of symmetry or inversion center and exists as non-superimposable mirror images (chiral enantiomers).
- 5 **A** Option A is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 6 **D** Option D is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 7 **B** Option B is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 8 **C** Option C is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 9 **B** Option B is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 10 **A** Option A is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 11 **D** Option D is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 12 **C** Option C is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 13 **A** Option A is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 14 **C** Option C is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 15 **B** Option B is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 16 **D** Option D is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 17 **C** Option C is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 18 **A** Option A is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 19 **D** Option D is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 20 **B** Option B is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 21 **B** Option B is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 22 **D** Option D is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 23 **A** Option A is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 24 **C** Option C is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 25 **A** Option A is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.
- 26 **D** Option D is correct. EDTA⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.

[illegible]

- MENTORA ACADEMY · Urwah · A Level Multiple Choice Pack — Topic 28 — Chemistry of Transition Elements (A Level MCQs)

81	B	Option B is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
82	D	Option D is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
83	A	Option A is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
84	C	Option C is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
85	A	Option A is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
86	D	Option D is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
87	B	Option B is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
88	C	Option C is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
89	B	Option B is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
90	A	Option A is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
91	D	Option D is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
92	C	Option C is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
93	A	Option A is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
94	C	Option C is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
95	B	Option B is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
96	D	Option D is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
97	C	Option C is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
98	A	Option A is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
99	D	Option D is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
100	B	Option B is correct. EDTA ⁴⁻ is a hexadentate ligand with two amine nitrogen atoms and four carboxylate oxygen atoms, all coordinating simultaneously to the central metal ion to give a coordination number of 6.	ting
101	C	Option C is correct. Displacing multiple monodentate ligands (e.g. 6 H ₂ O) with fewer multidentate ligands (e.g. 3 en or 1 EDTA) produces a net increase in independent molecules, generating a large favorable positive S°.	ces
102	A	Option A is correct. Displacing multiple monodentate ligands (e.g. 6 H ₂ O) with fewer multidentate ligands (e.g. 3 en or 1 EDTA) produces a net increase in independent molecules, generating a large favorable positive S°.	ces
103	D	Option D is correct. Displacing multiple monodentate ligands (e.g. 6 H ₂ O) with fewer multidentate ligands (e.g. 3 en or 1 EDTA) produces a net increase in independent molecules, generating a large favorable positive S°.	ces
104	B	Option B is correct. Displacing multiple monodentate ligands (e.g. 6 H ₂ O) with fewer multidentate ligands (e.g. 3 en or 1 EDTA) produces a net increase in independent molecules, generating a large favorable positive S°.	ces
105	A	Option A is correct. Displacing multiple monodentate ligands (e.g. 6 H ₂ O) with fewer multidentate ligands (e.g. 3 en or 1 EDTA) produces a net increase in independent molecules, generating a large favorable positive S°.	ces
106	C	Option C is correct. Displacing multiple monodentate ligands (e.g. 6 H ₂ O) with fewer multidentate ligands (e.g. 3 en or 1 EDTA) produces a net increase in independent molecules, generating a large favorable positive S°.	ces
107	B	Option B is correct. Displacing multiple monodentate ligands (e.g. 6 H ₂ O) with fewer multidentate ligands (e.g. 3 en or 1 EDTA) produces a net increase in independent molecules, generating a large favorable positive S°.	ces

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- 108 **D** Option D is correct. Displacing multiple monodentate ligands (e.g. 6 H₂O) with fewer multidentate ligands (e.g. 3 en or 1 EDTA) produces a net increase in independent molecules, generating a large favorable positive S°.
-
- 109 **A** Option A is correct. Displacing multiple monodentate ligands (e.g. 6 H₂O) with fewer multidentate ligands (e.g. 3 en or 1 EDTA) produces a net increase in independent molecules, generating a large favorable positive S°.
-
- 110 **C** Option C is correct. Displacing multiple monodentate ligands (e.g. 6 H₂O) with fewer multidentate ligands (e.g. 3 en or 1 EDTA) produces a net increase in independent molecules, generating a large favorable positive S°.
-